Katherine Deborah

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EDUCATION

University of Southern California (USC) December 2026
Master of Science in Computer Science *Los Angeles, California, USA*
Coursework: Fundamentals of Artificial Intelligence, Web Technology, Analysis of Algorithms

Government Institute of Technology June 2024
Bachelor of Technology in Information Technology *Coimbatore, India*

TECHNICAL SKILLS

Programming Language: Python, C++, Java, MATLAB, SQL, C, JavaScript, HTML, CSS
Libraries and frameworks: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, PyTorch, TensorFlow, Flask, Vue.js, React.js, Next.js, Node.js, Three.js, Streamlit
Tools and Platforms: Docker, Apache Kafka, Git, Tableau, Rest API
Databases: MySQL, PostgreSQL, MongoDB

WORK EXPERIENCE

Department of Telecommunications, Government of India Aug 2023-Dec 2023
Data Analyst *Coimbatore, India*

- Developed a full-stack web application to streamline inquiry responses using React (frontend), Flask (backend), and SQLite, integrating key insights from SIM card data to enable real-time, accurate lookups.
- Automated the previously manual inquiry process, cutting response time by over 10 hours through dynamic routing and database integration.
- Engineered backend algorithms to auto-route electronic files and track case status, improving departmental efficiency and transparency.
- Cleaned and processed 60M+ rows of SIM card metadata (80 fields) using Python, Pandas, and Scikit-learn; conducted anomaly detection and contributed a time-saving solution to recurring data inconsistencies.

PROJECTS

Influencer Engagement & Sponsorship Coordination Platform

- Created an Influencer Engagement & Sponsorship Coordination Platform deploying Flask, Vue.js, and SQLite, featuring role-based access for Admin, Sponsors, and Influencers to manage campaigns, ad requests, and profiles
- Enhanced functionality with Redis caching, Celery for automated tasks (reminders, reports), secure authentication via Flask-Security, and data export and visualization operating Flask-Excel and Chart.js

EduBridge

- Developed a full-stack web application using Next.js, React, Tailwind CSS, Node.js, and PostgreSQL to streamline learning progress tracking for teachers, with secure role-based authentication and middleware
- Implemented interactive dashboards enabling CSV export, report generation, and bulk email reminders directly from the UI. Designed responsive teacher and student views with session timelines, auto-saving journals, and real-time progress tracking

Video Analysis Q&A System

- Built a Streamlit-based UI for video uploads, extracting frames at 0.2 FPS and generating scene captions using the BLIP model vision-language LLM; stored captions in SQLite for efficient retrieval
- Integrated Whisper for speech-to-text transcription and Qwen LLM for natural language question answering, enabling users to query and receive context-aware responses about video content

CERTIFICATIONS

Diploma in Programming and Data Science – *IIT Madras*, Mathematics for Machine Learning - *Coursera Imperial College London,* Cloud Computing, Block Chain - *NPTEL IIT Kharagpur*